Indian Institute of Technology (Banaras Hindu University), Varanasi, U.P.

Mid Semester Examination, Sept' 2025

Duration:	: 02 Hours	Maximum marks	: 30
Programme	: M.Tech. Mining Engg. (Pt-I) and Ph.D	Semester	: Odd
Course Title	: Reliability Engineering	Course Code	: MN572

Note:

1) *Answer all questions.*
2) *Marks are indicated against each question in the brackets.*
3) *Take* $Z_{0.05} = \mp 1.645$; $Z_{0.12} = \mp 1.175$ and $t_{0.05,9} = 1.833$ *where applicable*

Q. 1 (a) Define and explain the terms 'Reliability' and 'Maintainability'. [2]

(b) Discuss mode (s) and cause(s) of failures with suitable examples. [2]

(c) What are the various methods for improving system reliability? [1]

(d) What are 'probability mass function' and 'probability density function'? [1]

(e) If $T \geq 0$ is a continuous random variable that represents the time to failure of a system, then give the mathematical expressions for the reliability function $R(t)$, cumulative distribution function $F(t)$, probability density function $f(t)$ and failure rate function $\lambda(t)$ at time t. Establish a relationship among these functions. [4]

Q. 2 (a) The Company 'X' manufactures chipset for use on smartphone. The time to failure, in year, of this chipset has the following PDF (Probability Density Function) [5]

$$f(t) = \frac{200}{(t+10)^3} \qquad for\ t \geq 0$$

(i) Determine the reliability of the chipset after first year of operation.
(ii) Compute the MTTF.
(iii) What is the design life for a reliability of 0.95?
(iv) Will a one-year burn-in period improve the reliability in part (i)? If so, what is the new reliability?

(b) A household appliance is advertised as having more than a 10-yr life. The PDF of the appliance is given by [3]

$$f(t) = 0.1(0.05t + 1)^{-3} \quad for\ t\ (in\ years) \geq 0$$

If it has survived a 1-year warranty period, what is its MTTF after the warranty period?

(c) The time to failure of an electric bulb follows a Weibull distribution with scale parameter of 2000 hours and shape parameter of 0.5. What is the probability that the electric bulb will last more than 4000 hours? What is the mean time to failure? [2]

Continued.....

Q.3 (a) The failure of a critical component of mining dozer is normally distributed with 90% of the failure occurring between 200 hours and 270 hours of operation, meaning 5% failure occurs before 200 hours of operation and 5% failure occurs after 270 hours of operation. [5]

(i) Calculate MTTF and σ of failure times of this component.
(ii) What is the reliability at the time of replacement when the component is replaced after 210 hr. of use?
(iii) Determine the design life if not more than a 1% probability of failure prior to the replacement is tolerated.
(iv) Estimate the reliability for another 10 hours of use if the component has been operating for 200 hours.

(b) Given the following 10 failures times in hours of an item X; [5]

24.5, 18.9, 54.7, 48.2, 20.1, 29.3, 15.4, 33.9, 72.0, 86.1.

Estimate $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$ of the component X at different time intervals for this ungrouped failure data set and compute a 90% confidence interval for the MTTF of the item X.

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M.Tech Part I/PhD Mining Engineering
Mid-Term Examination, 2025
MN573: Safety & Risk Management in Mines

Time: 2 Hrs. **Marks: 20**

1. Discuss how Heinrich's Safety Philosophy is a phenomenal departure from the then prevailing safety concept. (4)
   Using the concept of 'Domino Game' elaborate the injury sequence. What was his emphasis on prevention of accidents at workplaces?

2. Explain the process of risk management. Explain the process of risk assessment using qualitative methods developing a risk matrix. (4)

3. Explain various indicators used for measuring safety performance in mines and other industrial sectors with their relative merits and demerits. (4)

4a. What is a random variable? Can the sample mean ever be exactly equal to the population mean?–Under what circumstances? (2)

4b. Derive an expression for the standard deviation of a binomial random variable with parameters n and p. (2)

5a. A probability density function (PDF) is given as: (2)

$$f(x) = \begin{cases} \frac{3}{8}x^2; & 0 \leq n \leq 2 \\ 0; & otherwise \end{cases}$$

(i) Verify that this is a valid PDF.

(ii) Find P $(1 \leq X \leq 2)$.

5b. (i) What do you understand by safety? Is there any relationship between safety and risk. (1)

(ii) In an underground coal mine, the accident statistics for the year 2023 revealed that there were two fatalities, 5 serious bodily injuries and 14 reportable injuries. The coal mine operates 310 days in a year, three shifts in a day and eight hours in a shift. The average employment in a shift was 400 and the production of the mine was 0.85 million tons. Calculate serious bodily injury rate per million tons output. (1)

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M.Tech. PT. I (SEM. I) and Ph.D., MINING ENGINEERING
SUBJECT – MN 535: FINITE ELEMENT METHODS FOR GEOMECHANICS

Mid Semester Examination - 2025

Duration: 2 Hours **Max. Marks: 30**

Note: Answer all questions. Assume any data if required.

Q. 1 Solve the following simultaneous equations through any iterative method **(up to three iterations).** **[4]**

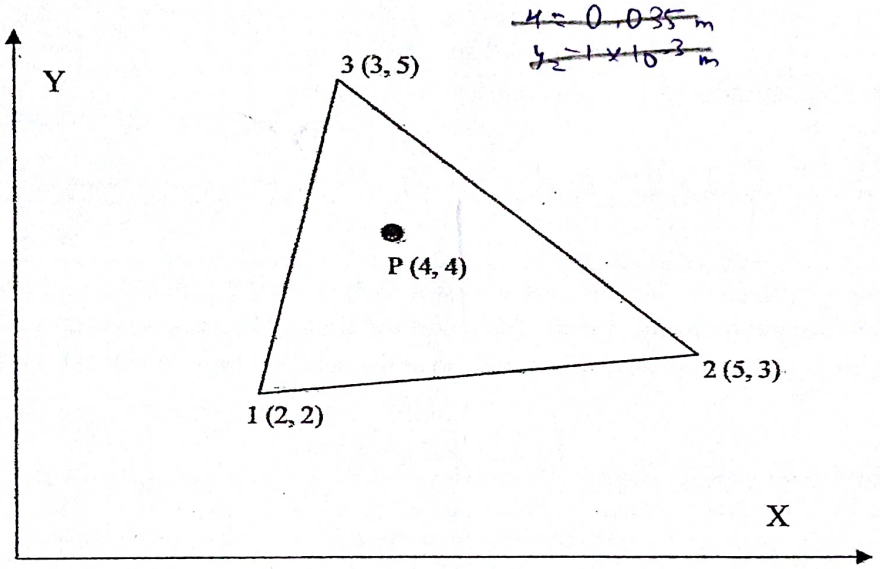
$2x + 3y + z = 10$
$x + 2y + 4z = 12$
$4x + y + 2z = -5$

$x = -3.14$
$y = 5$
$z = 1.2857$

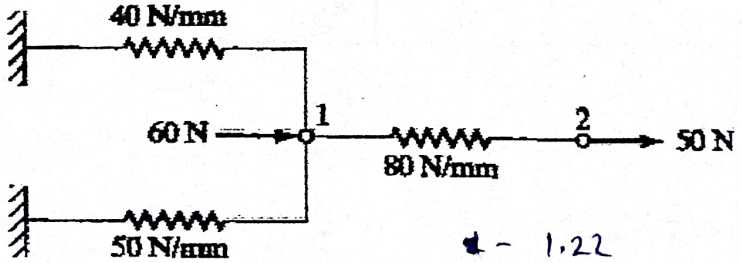
Q. 2 A triangular element is shown in the figure. The coordinates of the nodal points are given in **meters**. Calculate the **displacements** (in X and Y directions) and **strain** at point **P**. The value of displacement at nodes points in X and Y directions are given as: **[8]**

Direction of displacement in **meters**	Node no. 1	Node no. 2	Node no. 3
X	0.001	0.003	0.006
Y	-0.004	-0.001	0.002

~~$x = 0.035$ m~~
~~$y_2 = 1 \times 10^{-3}$ m~~



Q. 3 Calculate the displacement at node 1 and node 2 of the spring system by using finite element method. Stiffness of the springs and forces on the nodal points are given in the figure. **[7]**

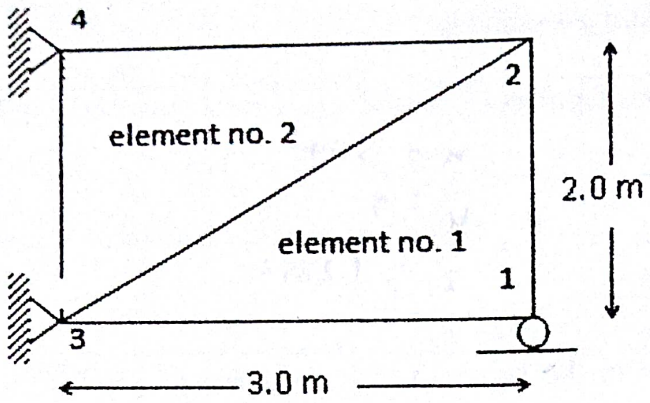


1 - 1.22
2 - 1.84

[P.T.O.]

Q.4

The local stiffness matrix (in N/m) of the given triangular shape **element no. 1**(shown in the figure), considering nodal numbers are given below. Determine the global stiffness matrix consists of both the elements no 1 and 2.

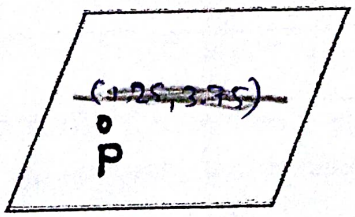


$$k = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0.983 & -0.5 & -0.45 & 0.2 & -0.533 & 0.3 \\ & 1.4 & 0.3 & -1.2 & 0.2 & -0.2 \\ & & 0.45 & 0 & 0 & -0.3 \\ & & & 1.2 & -0.2 & 0 \\ & \text{Symmetric} & & & 0.533 & 0 \\ & & & & & 0.2 \end{bmatrix} \end{array}$$

Q.5

A four point quadrilateral finite element is shown in the figure. Assume the coordinates of the nodal points in x-y coordinates system. Transform the element into the iso-paramtric element in ξ - η coordinate system. Calculate the coordinate of the point **P in x-y coordinate system at:** [4]

$$\xi = 0.25 \text{ and } \eta = 0.75.$$



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M.Tech/PhD I[st] Year Semester I Mid Semester Examination, Session 2025-26 Mining Engineering

Subject: Operations Research in Mining, Subject Code: MN561

Time: Two hours **Full Marks: 30**

Answer all questions.

1. Consider a problem with two decision variables, x_1 and x_2, which represent the levels of activities 1 and 2, respectively. For each variable, the permissible values are 0, 1, and 2, where the feasible combinations of these values for the two variables are determined from a variety of constraints. The objective is to maximize a certain measure of performance denoted by Z. The values of Z for the possibly feasible values of (x_1, x_2) are estimated to be those given in the following table: **[3]**

	Z	Z	Z
X_1	0	1	2
0	0	4	8
1	3	8	13
2	6	12	18

Based on this information, indicate whether this problem completely satisfies each of the four assumptions of linear programming. Justify your answers.

2. Consider the following problem. **[4]**

Maximize $Z = 3x_1 + 2x_2$

Subject to : $2x_1 + x_2 \leq 6$

$x_1 + 2x_2 \leq 6$

And $x_1 \geq 0$, $x_2 \geq 0$.

(a) Use the graphical method to solve this problem. Circle all the corner points on the graph.
(b) For each CPF solution, identify the pair of constraint boundary equations it satisfies.
(c) For each CPF solution, identify its adjacent CPF solutions.
(d) Calculate Z for each CPF solution. Use this information to identify an optimal solution.
(e) Describe graphically what the simplex method does step by step to solve the problem.

3. **[3]**

Determine the feasible space for each of the following independent constraints that $x_1, x_2 \geq 0$.

(a) $-3x_1 + x_2 \leq 6$.

(b) $x_1 - 2x_2 \geq 5$.

(c) $2x_1 - 3x_2 \leq 12$.

(d) $x_1 - x_2 \leq 0$.

(e) $-x_1 + x_2 \geq 0$.

4. The Omega Manufacturing Company has discontinued the production of a certain unprofitable product line. This act created considerable excess production capacity. Management is considering devoting this excess capacity to one or more of three products; call them products 1, 2, and 3. The available capacity on the machines that might limit output is summarized in the following table: [4]

Machine Type	Available Time (Machine Hours per Week)
Milling machine	500
Lathe	350
Grinder	150

The sales department indicates that the sales potential for products 1 and 2 exceeds the maximum production rate and that the sales potential for product 3 is 20 units per week. The unit profit would be \$50, \$20, and \$25, respectively, on products 1, 2, and 3. The objective is to determine how much of each product Omega should produce to maximize profit. Formulate a linear programming model for this problem. The number of machine hours required for each unit of the respective products is

Productivity coefficient (in machine hours per unit)			
Machine Type	Product 1	Product 2	Product 3
Milling machine	9	3	5
Lathe	5	4	0
Grinder	3	0	2

5. **Label each of the following statements about linear programming problems as true or false, and then justify your answer.** [3]

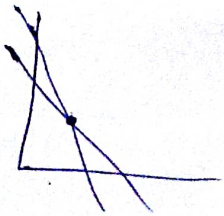
(a) For minimization problems, if the objective function evaluated at a CPF solution is no larger than its value at every adjacent CPF solution, then that solution is optimal.

(b) Only CPF solutions can be optimal, so the number of optimal solutions cannot exceed the number of CPF solutions.

(c) If multiple optimal solutions exist, then an optimal CPF solution may have an adjacent CPF solution that also is optimal (the same value of Z).

(d) The simplex method's rule for choosing the entering basic variable is used because it always leads to the best adjacent BF solution (largest Z).

(e) The simplex method's minimum ratio rule for choosing the leaving basic variable is used because making another choice with a larger ratio would yield a basic solution that is not feasible.



6. Consider the following set of constraints: [6]

$-2x_1 + 3x_2 = 3$ ----------(1)

$4x_1 + 5x_2 \geq 10$ ----------(2)

$x_1 + 2x_2 \leq 5$ ----------(3)

$6x_1 + 7x_2 \leq 3$ ----------(4)

$4x_1 + 8x_2 \geq 5$ ----------(5)

$x_1, x_2 \geq 0$

For each of the following problems, develop the z- row (any two : one maximization and one minimization) after substituting out the artificial variables:

(a) Maximize $z = 5x_1 + 6x_2$ subject to (1), (3), and (4).

(b) Maximize $z = 2x_1 - 7x_2$ subject to (1), (2), (4), and (5).

(c) Minimize $z = 3x_1 + 6x_2$ subject to (3), (4), and (5).

(d) Minimize $z = 4x_1 + 6x_2$ subject to (1), (2), and (5).

(e) Minimize $z = 3x_1 + 2x_2$ subject to (1) and (5).

7. Identify the direction of increase in z in each of the following cas [2]

(a) Maximize $z = x_1 - x_2$.

(b) Maximize $z = -5x_1 - 6x_2$.

(c) Maximize $z = -x_1 + 2x_2$.

(d) Maximize $z = -3x_1 + x_2$.

8. Ozark Farms uses at least 800 lb of special feed daily. The special feed is a mixture of corn and soybean meal with the following compositions: [5]

	lb per lb of feedstuff		
Feedstuff	*Protein*	*Fiber*	Cost ($/lb)
Corn	.09	.02	.30
Soybean meal	.60	.06	.90

The dietary requirements of the special feed are at least 30% protein and at most 5% fiber. Ozark Farms wishes to determine the daily minimum-cost feed mix.

With the help of graphical method detremine the minimum cost feed mix.

M.TECH / Ph.D.

COURSE CODE AND NAME: MN-552 – PROJECT MANAGEMENT

Mid Semester Examination (2025-26 - Odd Semester)

Duration - 2 Hrs

Total Marks - 30

Part A - Short Answers – Answer any 5

Question Number	Question	Marks
		[10]
1.	Define Project Life Cycle and list the key phases in the context of project management.	[1+1]
2.	Explain the different types of costs involved in mining projects.	[2]
3.	Differentiate between NPV and IRR.	[2]
4.	List the steps involved in the preparation of balance sheets as part of financial management.	[2]
5.	What is a project human resource management plan, and list the steps followed for developing HRMP.	[1+1]
6.	Explain the sources of Grievances?	[2]
7.	Differentiate between Induction and Training Programmes curated for Personnel Management in the mining industry.	[2]

Part B - Detailed Answers – Answer any 4

Question Number	Question	Marks
		[20]
1.	Explain in detail the project planning process, including its key components, steps, and tools used.	[1+1+2+1]
2.	What is Sensitivity Analysis, and detail the steps involved in conducting a sensitivity analysis for mining operations and financial management.	[2+3]
3.	What is the Recruitment Procedure followed by HR in Personnel Management System? Mention the steps followed to recruit and list the recruitment techniques commonly used in personnel recruitment.	[1+2+2]
4.	Explain the process of Grievance Handling Process in detail using a flowchart.	[5]
5.	Explain the types of organisational communication practiced in the mining industry.	[5]

********** All The Best **********

Indian Institute of Technology (BHU) Varanasi

Mid-Semester Examination — Odd Semester 2025-26

LLT-305: Literature, Culture, and Society

Marks: 30 **Time: 2 hours**

Section I

Answer any FIVE questions in about 500–700 words each. 5x5 = 25 Marks

1. "Literature adds to reality; it does not simply describe it. It enriches the necessary competencies that daily life requires and provides — in this respect, it irrigates the deserts that our lives have already become." Explain the relevance of studying literature.

2. Do an ecofeminist reading of A.K. Ramanujan's poem *Ecology*. How does the poem reflect the interconnectedness of nature, culture and gender?

3. In what ways does the river operate as a metaphor for the fragility of human civilisation in *The Ghaghra in Spate*? Discuss how Keki N. Daruwalla employs the motif of flood to interrogate the themes of transience, impermanence and the vulnerability of human settlements in the face of natural forces.

4. How does A. Revathi narrate her early experiences of gender dysphoria in her autobiography? Examine how the first chapter conveys the tension between childhood innocence and the growing awareness of being 'different.'

5. How can *Zero Sum Game* be interpreted as a critique of global power dynamics and postcolonial economic systems? Analyse how the essay employs the zero-sum framework and the use of satire to advance this critique.

6. Examine how the family in Shyam Selvadurai's *Funny Boy* functions as an institution for policing gendered behaviour. How do Arjie's parents and aunt respond to his play? What is the significance of the chapter title '*Pigs Can't Fly*' in relation to gender and familial expectations?

7. The essay *The Brand Expands* explores themes such as the homogenization of culture, the objectification of individuals as billboards and the exploitation of labour from the Global South. To what extent do you agree with this perspective? Discuss how multinational corporations use branding to dominate global markets, and evaluate the impact of these practices on local economies, labour conditions, and cultural diversity.

8. In *Still I Rise*, Maya Angelou not only speaks of personal resilience but also positions the poem within a larger cultural and historical struggle against racial and gender oppression. Discuss how the poem negotiates the tension between individual voice and collective memory.

9. In *Girls* by Mrinal Pande, the narrative repeatedly conveys the perception of daughters as burdens within the family. How does Pande critique this notion, and what does the story reveal about the socio-economic and cultural factors that sustain such views in patriarchal societies?

Section II

Write an annotation on any ONE of the following in about 500-700 words. 5x1 = 5 Marks

(You have to write the title and author of the text, the context in which the lines occur, and the relevance of the given lines in the text.) (2+1+2=5)

10. "Reading and writing, once started, would lead to play and novel writing, and household norms would be endangered. As he thought further about the matter, he worked out a most subtle theory. Perfect marriage was produced by a combination of female and male power. But if through women's education female power was weakened, then male power would prevail unchecked; and the clash between male and male would be so destructive that marriage would be annihilated, and women would be widowed. As yet, no one had been able to challenge this theory."

11. "Soon afterwards we left Baroda for Bombay
Where there are no trees except the one
Which grows and seethes in one's dreams, its aerial roots
Looking for the ground to strike."

Indian Institute of Technology (BHU) Varanasi

Mid-Term Examination (Odd Semester 2025-26)

MME- 306 (Media Environment)

arks: 30 Time: 2 hrs

Read the instructions carefully before answering.

Instructions:

- Section A Part 1: Attempt all 5 MCQs. Choose the one incorrect statement in each.
- Section A Part 2: Attempt all 5 short answer questions.
- Section B: Attempt any five out of seven questions.
- Section C: Attempt any two out of four questions. Use examples wherever relevant. Clarity of thought and structured responses will carry weight.

SECTION-A

Part-I

Identify the Incorrect Option. (5)

1. Which of the following accurately describes why Gen Z's reaction to smart glasses is important from a branding and market insight perspective?
   A) It shows Gen Z prefers more expensive technology over functionality.
   B) It demonstrates a generational rejection of all wearable tech.
   C) It signals that privacy-conscious consumers are influencing product design and adoption.
   D) It proves that TikTok is the primary platform for product launches.

2. What does Microsoft's development of in-house AI models (like MAI-1 and MAI-Voice-1) primarily indicate about the direction of tech branding?
   A) Companies are moving away from cloud to offline solutions.
   B) Big tech firms are building their own AI brands to reduce dependency and differentiate.
   C) Microsoft is divesting from the AI space.
   D) OpenAI is no longer relevant in the tech ecosystem.

3. What strategic branding insight can be drawn from Akasa Air's announcement to grow to 40 aircraft by 2026?
   A) It is planning to shift to cargo-only operations.
   B) It is positioning itself as a premium full-service airline.
   C) It is reinforcing its identity as a fast-scaling low-cost carrier targeting underserved markets.
   D) It is planning to exit the Indian market by 2026.

4. What does the Delhi Zoo closure due to bird flu tell us about media communication during public health incidents?
   A) Media should avoid discussing animal diseases to prevent panic.
   B) Recurrent coverage helps build public apathy toward health alerts.
   C) It illustrates how swift information dissemination protects public safety and drives risk perception.
   D) Media reporting is ineffective in influencing behavior during outbreaks.

5. The launch of India's first AI-driven "smart village" in Satnavari is an example of:
   A) Technological overreach with no social benefit.
   B) Government propaganda through digital media.
   C) Using media to highlight how technology can drive inclusive growth and inspire aspirational narratives.
   D) A corporate stunt with no long-term benefits.

Part-II

Short Conceptual Answers *(5 × 1 = 5 marks)*

1. Define the term "Emotional Value" in branding, with one Indian example.
2. What is the core difference between ATL and BTL advertising?
3. Name one key ethical concern with using AI in advertising.
4. What does the "Scarcity Bias" mean in behavioural economics?
5. In the Buyer Decision Process, what typically follows Information Search?

SECTION – B

[5 x 2]

Answer any FIVE of the following questions.

1. Explain the role of Print media in driving trust in Indian advertising.
2. How does Zomato use the AIDA framework in shaping its digital strategy?
3. Describe one consumer segment in India and how a brand tailors its campaign for it.
4. Discuss how Tanishq's Diwali Campaign successfully integrated multiple media touchpoints.
5. What is the "Job to be Done" for an urban Gen Z consumer buying Boat audio products?
6. Differentiate between functional and emotional brand drivers using Nykaa and Boat.
7. Explain the use of AI in enhancing advertising conversion, with one example.

SECTION – C

[2 x 5]

Answer any TWO of the following questions.

1. Case Analysis: Amul's Advertising Strategy
   Using the AIDA framework, break down how Amul sustains brand loyalty. Include cultural/media strategy elements.
2. Application: Buyer Decision Process
   Map a recent purchase (your own or someone else's) through the 5-stage decision process. Highlight biases and brand influences.
3. Case Study Interpretation: AVGC+ Studio
   Pick one customer segment and explain how advertising should be designed to convert this audience from Awareness to Purchase.
4. Critical Thinking: Resonance Over Reach
   "Advertising effectiveness is no longer just about reach, but about resonance." Discuss with examples from Indian campaigns.